Formalized Exterior-Model Spinors in Split Rank: Levi Actions, Non-Descent, and Exact Split-Line/Levi Images

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Abstract. We formalize in Lean 4/Mathlib an exterior-model spinor construction for split quadratic spaces over fields K with $2 \in K^\times$. The development keeps the chosen hyperbolic or split-Witt presentation as data, so the spinor module ΛW, its chiral halves, and the transported Clifford and spin actions remain available throughout the covering-map arguments. This explicit chosen model supports precise field-sensitive image theorems. In positive split rank the kernel of $\mathrm{Spin}(V, Q) \to \mathrm{SO}(V, Q)$ is $\{\pm 1\}$, and the exterior-model spin action therefore descends only projectively to the orthogonal image. For a split line, the spin image is exactly the square-scaling subgroup. In finite split rank with a distinguished line, a split Levi element lies in the spin image exactly when its determinant is a square in $K^\times$; in this finite-basis Levi setting, the determinant square-class character has the spin image as kernel and gives quotient isomorphisms to $K^\times/(K^\times)^2$. The formalization also gives explicit Clifford-unit representatives for transvections and chosen-line square scalings, their torus law, square-surjective field corollaries, and a Clifford-level vector-product norm substrate with order and repetition invariance.

Mathematics Subject Classification (2020). Primary 15A66; Secondary 03B35, 68V15.

Keywords. Clifford algebra, Formalized mathematics, Lean, Mathlib, Spin group, Spinor representation.

1. Introduction

For Clifford algebraists, the exterior-model construction of spinors is classical: after choosing a maximal totally isotropic subspace W inside a hyperbolic space, the spinor module is ΛW, Clifford generators act by wedge and contraction, and the even part produces the chiral halves. What is less stable over general fields is the bookkeeping of the chosen splitting and the exact scope of the spin covering map when nonsquare units are present.

We therefore keep this exterior-model construction explicit and make the covering and image statements sensitive to square classes in the base field.

We develop the classical construction of a spinor module $S = \Lambda W$ attached to an explicit hyperbolic presentation $Q \simeq \mathrm{dualProd}_K W$, transport it through Witt-style reformulations, and obtain the induced $\mathrm{Spin}(V, Q)$-action. Throughout, the spinor module is the exterior model ΛW supplied by a hyperbolic or split-Witt presentation; this is the object with the classical spin dimension and chiral decomposition.

In Lean 4/Mathlib, these statements require transported hyperbolic or Witt data and explicit Clifford calculations on the same exterior model. The resulting machine-checked theorems include the Levi-action bridge, the exact split-line image theorem, the finite-basis square-determinant factorization with its explicit Clifford lift, and the finite-basis split-Levi spin-image iff.

Contributions. Our main contribution is a reusable interface that transports Clifford and spin actions from explicit hyperbolic/Witt data to theorem-level covering statements. The three centerpiece mathematical outputs are the structural Levi-action theorem, the exact split-line image theorem, and the finite-basis square-determinant factorization together with its explicit Clifford lift and exact split-Levi spin-image criterion. Supporting these are the top-degree pairing and star-normalization lemmas that prove determinant-square necessity, the positive split-rank kernel/projective-descent results, and the explicit hyperbolic transvection Clifford units and chosen-line square-scaling lifts with their weight-2 torus law. The classical exterior-model equivalence and half-spin construction serves as infrastructure on which these field-sensitive theorems rest; the roadmap below points to the detailed theorem statements.

Standing conventions. Unless explicitly specialized to $\mathbb{R}$ or $\mathbb{C}$, the paper works over a field K with $2 \in K^\times$. The notation $\mathrm{dualProd}_K W$ always denotes the standard hyperbolic form on $W^* \times W$. Chosen exterior-model theorems are stated either for an explicit hyperbolic presentation $e : Q \simeq \mathrm{dualProd}_K W$ or, in the split-rank reformulation obtained from a chosen Witt decomposition, for a finite-dimensional nondegenerate form Q with $\dim V = 2\,\mathrm{wittIndex}(Q)$. Statements about the negative half-spin module beyond its formal zero-rank occurrence, and statements about equal half-spin dimensions, additionally assume positive split rank. Here a split-Witt presentation means the hyperbolic presentation induced by a chosen Witt decomposition. The vacuum line in an exterior model is the line $K \cdot 1 \subseteq \Lambda W$. We write $H(W)$ for the hyperbolic form $\mathrm{dualProd}_K W$. For a linear automorphism g of W, $g^\vee$ denotes the induced dual map on W^*, and $g^{-\vee}$ means $(g^{-1})^\vee$. We write $\iota_Q(v)$, or simply $\iota(v)$ when Q is clear, for the canonical Clifford generator attached to the vector v. The notation ρ_S denotes the transported Clifford action on the chosen exterior model $S = \Lambda W$, and $\mathrm{Ext}(g)$ denotes the exterior action induced by a linear map g on W. We write $\tilde{x}$ for Clifford conjugation (grade involution followed by reversion), so $\tilde{v} = -v$ for vectors, and the Clifford conjugate of a product vw of orthogonal vectors is $-vw$. We write $x^\star$ for reversion in the top-pairing argument below; on even spin elements used there, $x^\star = \tilde{x}$. The group $\mathrm{Spin}(V, Q)$ is the even subgroup of the Clifford units generated by products of unit vectors. In particular, any product of two vectors of square ± 1 lies in $\mathrm{Spin}(V, Q)$, and every $x \in \mathrm{Spin}(V, Q)$ satisfies $x\tilde{x} = 1$. All statements about the map $\mathrm{Spin}(V, Q) \to \mathrm{SO}(V, Q)$ concern the corresponding groups of K-points; no stronger scheme-theoretic claim is intended.

Formalized result boundary. The Lean development verifies the theorems stated here. Its central image theorem is exact on the finite-basis split Levi subgroup: spin-image membership is equivalent to the determinant being a square. The corresponding determinant square-class homomorphism is onto and has the spin image as kernel, both in linear coordinates and after transport to the canonical Levi subgroup. These kernel statements are also expressed as quotient isomorphisms by the first isomorphism theorem, and the same local split-Levi characters are exposed under spinor-norm-facing wrapper names. The square-surjective corollary removes the determinant-square side condition for all Levi elements.

The supporting Lean files also prove the chosen exterior-model spinor interface, the split-rank kernel/projective-descent results, explicit transvection and chosen-line square-scaling lifts, the Levi-projective exterior-action bridge, and the exact split-line image criterion. In addition, it provides a Clifford-level vector-product norm formula with invertible-vector square-class invariants, product-order invariance, repeated-pair cancellation, and duplicated-product/subproduct square-class triviality.

The following topics are outside the scope of the present paper: full Bott-period-8 real classification beyond the packaged rows through $\mathrm{Cl}(0, 8)$ and the positive even companions through $\mathrm{Cl}^+(8, 0)$; full all-orthogonal-group image classification; global orthogonal-group spinor-norm theory beyond the product-level substrate; and scheme-theoretic spin-cover statements.

The split line is the smallest test case. When $\dim W = 1$, the exact image theorem shows that $\mathrm{Spin}(H(W)) \to \mathrm{SO}(H(W))$ hits precisely the square-scaling subgroup, so the usual split-line double-cover slogan is correct only under square-surjectivity hypotheses on $K^\times$. The finite-basis Levi results proved below are the structural and constructive extension of that field-sensitive rank-1 picture: square determinant forces a concrete chosen exterior-model lift with normalized exterior action and actual membership in the spin image, while the same top-degree pairing normalization proves the converse determinant-square obstruction.

2. Context

Mathematical background. Our treatment follows the classical presentations of Lawson–Michelsohn [5], Chevalley [4], Atiyah–Bott–Shapiro [2], and Lounesto [6], with the low-dimensional octonionic and quaternionic incidences drawn from Baez [3]. The underlying “choose a maximal isotropic and use

ΛW" construction is classical; the point here is to keep the choice data explicit and stable under transport.

Formalization context and prior work. Lean 4 itself is described by de Moura and Ullrich [7]. Mathlib already provides the ambient Clifford and exterior algebra libraries, including `CliffordAlgebra`, `ExteriorAlgebra`, `pinGroup`, and `spinGroup` [8, 9]. Wieser and Song's Lean geometric-algebra formalization [11] and the earlier Lean 3 `lean-ga` library [10] supplied precursor infrastructure for Clifford-style reasoning. The present work builds a chosen exterior-model spinor layer over the current Lean 4/Mathlib stack: first-class hyperbolic/Witt presentations, transported exterior-model spin actions, explicit root/torus elements in the split model, and image theorems stated directly on the resulting chosen model. It combines these transported spinor actions with the Levi-projective theorem, the exact split-line image theorem, and the finite-basis square-determinant factorization, including its explicit Clifford lift, exact spin-image iff, and square-surjective all-Levi corollaries.

Novelty boundary. We treat the exterior-model realization $\mathrm{Cl}(Q) \simeq \mathrm{End}(\Lambda W)$, the parity split, and the basic chosen exterior-model reconstruction of spinors as classical. The new content is the field-sensitive covering/image behavior in the transported exterior model: the positive split-rank linear/projective dichotomy; the explicit transvection Clifford units and chosen-line square-scaling lifts; their internal and orthogonal weight-2 semidirect structure together with the exact normalized exterior action of the torus lift; the theorem that split Levi lifts act projectively as the natural exterior action; the finite-basis square-determinant factorization and its explicit even unitary Clifford lift; the determinant-square necessity theorem, resulting exact finite-basis Levi image criterion, and determinant square-class quotient character with its surjectivity, kernel theorem, and quotient-isomorphism form, together with local split-Levi spinor-norm-facing wrappers for the same character; the square-surjective all-Levi corollary; and the exact split-line image theorem. The common thread is a concrete chosen exterior-model realization of a root subgroup, its square torus, and the resulting square-class obstruction.

Results roadmap. The detailed statements appear in Section 5: the structural Levi-action theorem, the exact split-line image theorem, the constructive square-determinant Levi factorization, and the exact finite-basis split-Levi spin-image criterion. Section 6 then gives the explicit Clifford representatives and proof ingredients that support those results.

3. Artifact and reproducibility

Repository	`https://github.com/Arthur742Ramos/SpinorLean`
SpinorLean code commit	`e7c7bf09e6c4477124e85a7d4339c0b4a97e52ca`
Lean toolchain	`leanprover/lean4:v4.30.0-rc1`
Mathlib commit	`11873348d8bf19440253d9282cc4e9423432e5ea`
Build command	`lake build`
Lean files under `Spinor/`	26
Physical Lean LOC under `Spinor/` (`wc -l`)	22,948
`sorry` occurrences under `Spinor/`	0

Artifact status. The Lean development proves the formal results stated above. The theorem claims in Sections 5 and 6 are tied to named Lean declarations in `THEOREM_INDEX.md`; the companion module `Spinor.TheoremIndex` re-exports and checks those declarations, and `Spinor.lean` imports that module so a normal build catches stale theorem names. The reproducible artifact path is: use the pinned `lean-toolchain` and `lake-manifest.json`, optionally fetch cached dependencies with `lake exe cache get`, run `lake build`, and run the proof-hole audit in `scripts/verify.sh` or `scripts/verify.ps1`. The paper source itself builds from `paper/main.tex` with the bibliography in `paper/refs.bib`.

The files most directly supporting the present paper are `Presentation.lean`, `ExteriorModel.lean`, `HyperbolicAction.lean`, `OrthogonalAction.lean`, `OddClassification.lean`, and `CliffordNorm.lean`; together they provide the chosen exterior-model vacuum-line lemmas, the structural Levi-action theorem, the explicit transvection Clifford units and square-scaling lifts with their conjugation laws, the finite-basis square-determinant Levi factorization, explicit Clifford lift, spin-image sufficiency and necessity with its square-surjective corollary, the top-degree pairing normalization underlying the necessity proof, the exact split-line image theorem, and the Clifford vector-product norm formula underlying reflection-product spinor norms.

The reusable interface is organized around the chosen exterior-model transport story. `Presentation.lean` packages explicit hyperbolic presentations as first-class data carrying the chosen model ΛW, its even/odd halves, the transported Clifford and spin actions, and the corresponding Witt-index facts. On top of that layer, `CliffordAction.lean`, `SpinRep.lean`, and `Chiral.lean` expose the transported actions and parity pieces as reusable algebra/module structures, while `OrthogonalAction.lean` packages the spin-to-isometry homomorphism and the explicit root/torus elements used in the later covering statements. This is how the Lean architecture mirrors the paper: the same chosen model is reused from the algebraic foundation all the way to the field-sensitive image theorems. Subsequent developments can therefore import these APIs directly, without reopening the basis-level wedge/contraction construction each time a new covering or representation statement is needed.

For auditability, the repository also contains `THEOREM_INDEX.md`, a reader-facing map from the paper's theorem labels to the Lean declarations that implement or support them.

Reusable interfaces. The reusable payoff is organized in four layers, each exposed as a small Lean interface.

Chosen exterior-model transport. Lean interface: the presentation, Clifford-action, and spin-representation modules. Later developments can state spinor theorems directly on a distinguished exterior model without redoing the split-model construction; exact module and declaration names are listed in `THEOREM_INDEX.md`.

Levi-projective exterior action. Lean interface: the scalar-vacuum theorem and the split Clifford action theorem indexed in `THEOREM_INDEX.md`. Together they provide the bridge from spin lifts to `ExteriorAlgebra.map`, with the scalar recorded as a unit.

Exact split-line image and finite-basis Levi image. Lean interface: the split-line image theorem, the factorization theorems in `OrthogonalAction.lean`, and the determinant-square necessity theorem in `HyperbolicAction.lean`. This exposes theorem-level image results and explicit root/torus generators for later extensions of the Levi criterion.

Clifford product norm substrate. Lean interface: `CliffordNorm.lean`, with exact declaration names indexed in `THEOREM_INDEX.md`. This records the global Clifford calculation for products of vector generators and packages the corresponding square-class invariant for products of invertible vectors. The scalar, unit, and square-class products are proved invariant under permutations and reversal of the chosen vector list; inserting a repeated invertible vector pair, duplicating a whole invertible-vector product or contiguous subproduct, and repeating a subproduct with intervening factors are also proved square-class-trivial, including after first permuting a list into the repeated form. Decomposition-independence across different Cartan–Dieudonne factorizations and the resulting orthogonal-group spinor norm remain later work.

Downstream reuse already present in the repository. The interface is already reused beyond the core covering-map story in the low-dimensional identification layer. Separate modules package the split, low-rank, and Bott-table entries through the negative period-eight row and the corresponding positive even companions. These supporting rows include real, complex, quaternionic, and product-matrix identifications for dimensions through eight. They show that the transport and orthogonal-action APIs are already sufficient for later formal developments connecting the split-model infrastructure to classical compact spin identifications. The exact module and theorem names for these rows are recorded in `THEOREM_INDEX.md`.

Gap relative to existing Mathlib infrastructure. The ambient Mathlib Clifford/exterior infrastructure is essential, but by itself it does not supply a distinguished exterior-model spinor module attached to a chosen maximal isotropic, nor does it transport that chosen model through Witt data to theorem-level covering statements. The main mechanized obstacle overcome here is precisely that interface gap: the Levi-projective theorem and the exact split-line image theorem both require one to compare Clifford conjugation, contraction/wedge operators, and orthogonal actions on the same transported exterior model. They do not follow from the abstract `spinGroup`/`pinGroup` API alone, because their content is about a specific chosen exterior-model realization of spinors and the field-sensitive image data visible there.

The proof-engineering obligations are structural. In `ExteriorModel.lean`, the vacuum-line criterion had to be stated and proved in a form stable under later transport: if every contraction kills a vector, then that vector lies on the vacuum line, and any endomorphism sending the vacuum to a scalar while intertwining wedge operators is already a scalar multiple of the exterior functor. In `OddClassification.lean`, the exact split-line image theorem requires an explicit even-part description of $\mathrm{Cl}^+(H(K))$ and a direct conjugation calculation inside that basis. In `OrthogonalAction.lean`, the higher-rank factorization bridges Mathlib's matrix theorem `Matrix.Pivot.exists_list_transvec_mul_diagonal_mul_list_transvec` to transported basis transvections and the canonical determinant-one 2×2 blocks. These constructions package the transport and covering-map arguments into reusable Lean interfaces.

4. Chosen exterior-model interface

This section records the classical chosen exterior-model infrastructure used later, but it also highlights the formal interfaces through which the Lean development reuses that setup. Readers interested primarily in the field-sensitive covering statements can skim to Section 5 and return here for the transported hyperbolic/Witt layer.

Definition 4.1 (Hyperbolic presentation and chosen spinor model). A hyperbolic presentation of a finite-dimensional quadratic form Q over K is an isometry $Q \simeq \mathrm{dualProd}_K W$ with W finite-dimensional, where $\mathrm{dualProd}_K W$ denotes the standard hyperbolic form on $W^* \times W$. The associated chosen spinor model is $S(W) = \Lambda W$, with parity pieces $S^{\pm}(W) = \Lambda^{\mathrm{even}/\mathrm{odd}} W$.

Hyperbolic presentation and transport. The central organizing idea is to isolate the data

$$Q \simeq \mathrm{dualProd}_K W \qquad (W \text{ a maximal totally isotropic subspace})$$

as a specified choice of maximal isotropic subspace together with an isometry to the standard hyperbolic form. From that single piece of data one obtains the chosen exterior model ΛW, its even and odd halves, the transported Clifford action $\mathrm{Cl}(Q) \to \mathrm{End}(\Lambda W)$, and the restricted spin-group action. The decisive point is that our "spinor module" is not the ambient regular module on V, but a distinguished exterior model carried along with the presentation data that justifies it.

The construction then proceeds in three steps. First, the split model on $W^* \times W$ is handled concretely, with wedge and contraction giving the Clifford action and explicit basis projectors giving $\mathrm{Cl}(W^* \times W, \mathrm{dualProd}) \simeq \mathrm{End}(\Lambda W)$. Second, any hyperbolic presentation transports that concrete result to the target quadratic space and its spin action. Third, Witt decomposition and the split-rank reformulation recover top-level statements that no longer require users to carry an explicit isometry argument. In this way, each substantive theorem is proved once on the concrete model and then reused at the levels where classification and orthogonal-action results are stated.

Representative Lean declarations. Our main theorems are mirrored directly by reusable Lean declarations. For example, the theorem index records declarations for the scalar-vacuum theorem, the split Clifford action theorem, and the split-line image theorem. These entries illustrate the design goal: theorem-level covering statements are exposed directly at the Lean interface, not only as consequences hidden behind a large background equivalence.

5. Main Results

Throughout this section, K is a field with $2 \in K^\times$ and Q is a finite-dimensional quadratic form. The algebra equivalences are first stated for an explicit hyperbolic presentation $e : Q \simeq \mathrm{dualProd}_K W$ with W finite-dimensional; their split-rank reformulations additionally assume that Q is nondegenerate and $\dim V = 2\,\mathrm{wittIndex}(Q)$. Statements involving a nontrivial negative half or the equal-dimension formulas additionally assume $\dim W > 0$ (equivalently positive split rank in the split-rank reformulation). The kernel theorem uses the finite-dimensional nondegenerate split-rank setting.

Classical chosen exterior-model foundation

The first theorem records the standard exterior-model material that the later covering and image statements use.

Theorem 5.1 (Exterior-model spinor theorem). *Assume K is a field with $2 \in K^\times$, let $e : Q \simeq \mathrm{dualProd}_K W$ be an explicit hyperbolic presentation, and set $S = \Lambda W$, $S^+ = \Lambda^{\mathrm{even}} W$, and $S^- = \Lambda^{\mathrm{odd}} W$. Then the following hold.*

1. *For the explicit hyperbolic presentation, the Clifford action on S by wedge and contraction yields*
$$\mathrm{Cl}(Q) \simeq \mathrm{End}(S).$$
If $\dim W > 0$, then the even action yields
$$\mathrm{Cl}^+(Q) \simeq \mathrm{End}(S^+) \times \mathrm{End}(S^-);$$
if $\dim W = 0$, then $\mathrm{Cl}^+(Q) = K = \mathrm{End}(S^+)$ and $S^- = 0$.
2. *If $\dim W = n$, then $\dim S = 2^n$; if $n > 0$, then $\dim S^+ = \dim S^- = 2^{n-1}$.*
3. *For finite-dimensional nondegenerate split-rank Q, any chosen Witt decomposition produces a hyperbolic presentation and hence a spinor model with the preceding properties. The positive half-spin module is simple for each such choice. In positive split rank the negative half-spin module is also simple and inequivalent to the positive half-spin module over the even Clifford algebra; in split rank 0, one has $S^+ = K$ and $S^- = 0$.*

Proof provenance. The explicit-presentation statement is obtained in Section 6 by transporting the split-model matrix-unit calculation (Theorem 6.2) and its parity corollary (Corollary 6.3) along Proposition 6.4. The dimension formulas are the standard binomial counts for ΛW, and the split-rank formulation follows by choosing a Witt decomposition and applying the same transported result. We state the result here because the later covering and image theorems are all built on this chosen exterior-model interface. □

We deliberately separate the presentation-dependent and split-rank versions of these results. The endomorphism-algebra equivalences are first proved from an explicit hyperbolic presentation, where no separate nondegeneracy hypothesis is needed; the nondegenerate split-rank assumptions enter when the explicit isometry data are suppressed and the chosen model is recast through a Witt construction.

At split rank 0, one simply has $W = 0$, $S^+ = K$, and $S^- = 0$. Positive-rank hypotheses first enter when one asks for a nontrivial negative half-spin module, for the equal half-spin dimension formula, or for the two-factor even-part decomposition.

This theorem is the algebraic engine behind the paper, but it is still classical chosen exterior-model setup. The new image results begin with the field-sensitive covering and image statements stated next.

New field-sensitive covering and image results

Theorem 5.2 (Positive split-rank kernel, spinorial descent, and exact split–line image). *Assume K is a field with $2 \in K^\times$. If Q is finite-dimensional, nondegenerate, and of positive split rank*

$$0 < \dim V = 2\,\mathrm{wittIndex}(Q),$$

then the kernel of the spin-to-isometry map

$$\mathrm{Spin}(V, Q) \to \mathrm{SO}(V, Q)$$

is exactly $\{\pm 1\}$. If $S = \Lambda W$ is the exterior-model spinor module attached to any chosen Witt decomposition of Q, then the resulting spin representation

$$\mathrm{Spin}(V, Q) \to \mathrm{GL}(S)$$

does not factor through $\mathrm{SO}(V, Q)$: there is no group homomorphism $\rho : \mathrm{SO}(V, Q) \to \mathrm{GL}(S)$ whose composite with $\mathrm{Spin}(V, Q) \to \mathrm{SO}(V, Q)$ is the given spin action on S. The same is true for each nonzero half-spin piece $S^\pm$. However, the induced action on the projective spaces $\mathbb{P}(S)$ and $\mathbb{P}(S^\pm)$ depends only on the image of $\mathrm{Spin}(V, Q) \to \mathrm{SO}(V, Q)$ and therefore defines an action of that image subgroup. For the split line $Q = \mathrm{dualProd}_K K$, the image inside $\mathrm{SO}(Q)$ is exactly the square-scaling subgroup, i.e. the subgroup of maps

$$(x, y) \longmapsto (ax, a^{-1}y) \qquad (a \in (K^\times)^2).$$

Consequently, the expected double-cover statement on the split line is recovered if and only if the square map on $K^\times$ is surjective; if $K^\times$ contains a nonsquare *unit, the split-line spin map is formally proved not to be surjective. No full all-orthogonal-group surjectivity theorem is claimed here.*

Proof provenance. Section 6 proves the kernel calculation (Proposition 6.5), non-factorization (Proposition 6.6), projective descent (Proposition 6.7), and the exact split-line image statement (Proposition 6.13). The final split-line clause is exact because every element of $\mathrm{SO}(\mathrm{dualProd}_K K)$ is a reciprocal scaling and Proposition 6.13 identifies precisely which such scalings lie in the spin image. $\square$

The next two statements are the paper's further headline results. Their proofs are deferred to the later section, after the explicit root and torus constructions needed for the argument have been set up.

Theorem 5.3 (Levi lifts act projectively as the exterior action). *Let $Q = \mathrm{dualProd}_K W$ and let $S = \Lambda W$ be the chosen exterior model. Suppose $s \in \mathrm{Spin}(Q)$ projects to the transported Levi element*

$$\Lambda(g) = (g^{-\vee}, g) \in \mathrm{SO}(Q)$$

for some $g \in \mathrm{GL}(W)$. Then there exists $c \in K^\times$ such that

$$\rho_S(s) = c\,\mathrm{Ext}(g)$$

as endomorphisms of S. In particular, the projective action of a split Levi lift on S depends only on its Levi coordinate g.

Theorem 5.4 (Square-determinant Levi factorization and exact spin-image criterion). *Let $S = \Lambda W$. Assume W has an ordered finite basis $(e_1, \dots, e_n)$ with $n \geq 1$, and take the line Ke_1 as distinguished. Let*

$$\Lambda : \mathrm{GL}(W) \hookrightarrow \mathrm{SO}(\mathrm{dualProd}_K W)$$

denote the transported Levi embedding $\Lambda(g) = (g^{-\vee}, g)$. If $g \in \mathrm{GL}(W)$ has $\det(g) = u^2$ for some $u \in K^\times$, then after choosing a basis of W and a distinguished basis line, $\Lambda(g)$ factors in $\mathrm{SO}(\mathrm{dualProd}_K W)$ as

$$\begin{aligned}\Lambda(g) = &\ \textit{(transported basis transvections)}\\ &\cdot \textit{(one chosen-line square scaling)}\\ &\cdot \textit{(canonical determinant-one } 2 \times 2 \textit{ diagonal blocks)}\\ &\cdot \textit{(transported basis transvections).}\end{aligned}$$

In particular, because each canonical 2×2 block is itself a product of transported hyperbolic transvections, $\Lambda(g)$ is a product of transported hyperbolic transvections and one chosen-line square scaling. Equivalently, the square-determinant part of the Levi subgroup admits explicit Clifford representatives implementing this factorization on $W^* \oplus W$. More concretely, one can choose an even unitary Clifford unit

$$x \in \mathrm{Cl}(\mathrm{dualProd}_K W)^\times,$$

built from those same explicit transvection and chosen-line factors, such that

$$\rho_S(x) = -u^{-1}\,\mathrm{Ext}(g)$$

as endomorphisms of S*. Consequently,* $\Lambda(\mathrm{SL}(W))$ *is contained in the subgroup of* $\mathrm{SO}(\mathrm{dualProd}_K W)$ *generated by the transported hyperbolic transvections. If the square map on* $K^\times$ *is surjective, the same Clifford-unit conclusion holds for every* $g \in \mathrm{GL}(W)$ *after choosing* $u \in K^\times$ *with* $\det(g) = u^2$*.*

The same square-determinant condition also gives actual spin-image membership:

$$\Lambda(g) \in \mathrm{im}(\mathrm{Spin}(\mathrm{dualProd}_K W) \to \mathrm{SO}(\mathrm{dualProd}_K W)).$$

Conversely, if $\Lambda(g)$ *lies in this spin image, then* $\det(g) = u^2$ *for some* $u \in K^\times$*. Hence, for finite-basis Levi elements with a distinguished basis line,* $\Lambda(g)$ *is in the spin image precisely when* $\det(g) \in (K^\times)^2$*. Equivalently, for the formalized homomorphism*

$$\chi : \mathrm{GL}(W) \longrightarrow K^\times/(K^\times)^2, \qquad \chi(g) = [\det(g)],$$

and for its transport to the canonical split Levi subgroup through $\mathrm{GL}(W) \simeq \Lambda(\mathrm{GL}(W))$*, the kernel of* χ *is exactly the pulled-back split-Levi spin image, and the transported kernel is exactly the canonical split-Levi spin image, as subgroup equalities. Moreover* χ *and its transported split-Levi character are surjective: every square class is represented by scaling the distinguished basis line. Therefore the first isomorphism theorem identifies both* $\mathrm{GL}(W)$ *modulo the pulled-back spin-image subgroup and the canonical split Levi subgroup modulo its spin-image subgroup with* $K^\times/(K^\times)^2$*. In particular, square-surjectivity of* $K^\times \xrightarrow{(-)^2} K^\times$ *implies that every finite-basis Levi element* $\Lambda(g)$ *lies in the spin image.*

Together, Theorems 5.1, 5.2, 5.3, and 5.4 capture our main arc: a classical exterior-model foundation, field-sensitive covering results, a structural Levi bridge, and an exact finite-basis Levi image theorem. The next section proves these statements and the supporting explicit calculations.

6. Mechanized proof ingredients

This section isolates the split-model calculations, transport lemmas, and explicit orthogonal-action formulas that underlie the theorem statements of Section 5. In the Lean development these are the reusable interfaces exported by the presentation and orthogonal-action files, so we present them here as the mechanized proof architecture rather than as a second main-results section.

Fix the split hyperbolic form $Q = \mathrm{dualProd}_K W$, choose a basis $e_1, \ldots, e_n$ of W with dual basis $e^1, \ldots, e^n$, and write e_I for the exterior basis vector indexed by a subset $I \subseteq \{1, \ldots, n\}$. Let $L_i(x) = e_i \wedge x$ and $D_i(x) = \iota_{e^i}(x)$ be wedge and contraction. These operators satisfy the split Clifford relations

$$L_i^2 = D_i^2 = 0, \qquad L_i D_j + D_j L_i = \delta_{ij}.$$

Lemma 6.1 (Projector factors on the exterior basis). *For a basis vector e_J and an index i, one has*

$$L_iD_i(e_J) = \begin{cases} e_J, & i \in J, \\ 0, & i \notin J, \end{cases} \qquad D_iL_i(e_J) = \begin{cases} e_J, & i \notin J, \\ 0, & i \in J. \end{cases}$$

Moreover, for distinct indices these operators commute on the exterior basis.

Proof. If $i \in J$, then $D_i(e_J)$ removes the basis vector e_i from e_J up to the usual Koszul sign, and wedging again by e_i restores e_J with the same sign. If $i \notin J$, then $D_i(e_J) = 0$. This proves the formula for L_iD_i; the formula for D_iL_i is identical, with "insert then remove" replacing "remove then insert." For distinct indices, the operators merely test membership of different basis vectors, so applying them in either order to any basis vector either reproduces that vector or kills it. Hence they commute on the exterior basis, and therefore on all of ΛW. □

Theorem 6.2 (Matrix units in the split model). *For the split hyperbolic form $Q = \mathrm{dualProd}_K W$, define for each subset $I \subseteq \{1, \dots, n\}$*

$$P_I = \prod_{i \in I} L_iD_i \prod_{j \notin I} D_jL_j.$$

In this formula, and in the one below, all products are taken from left to right in increasing index order. Then $P_I(e_J) = 0$ for $J \neq I$ and $P_I(e_I) = e_I$. For subsets I, J, let

$$T_{I,J} = \varepsilon_{I,J}(\prod_{i \in I \setminus J} L_i)(\prod_{j \in J \setminus I} D_j)P_J,$$

where $\varepsilon_{I,J} \in \{\pm 1\}$ is the unique sign making $T_{I,J}(e_J) = e_I$. Then

$$T_{I,J}(e_{J'}) = \delta_{J,J'}e_I.$$

Consequently the image of the split Clifford action contains every matrix unit $E_{I,J}$ of $\mathrm{End}(\Lambda W)$. Hence the split Clifford action is surjective, and since both source and target have dimension 2^{2n}, one obtains

$$\mathrm{Cl}(Q) \simeq \mathrm{End}(\Lambda W).$$

Proof. The preceding lemma shows that each factor in P_I either fixes or kills a basis vector according to whether the corresponding index belongs to that basis vector. Since the factors commute on the exterior basis, the product P_I annihilates every e_J with $J \neq I$ and fixes e_I. Thus P_I projects onto the line Ke_I.

Now fix J. The operator P_J kills every basis vector except e_J. On the vector e_J, the contraction factors indexed by $J \setminus I$ are applied in increasing order and remove exactly the basis elements that must disappear; the result is the basis vector $e_{J \cap I}$ up to the corresponding Koszul sign. The wedge factors indexed by $I \setminus J$, again in increasing order, insert exactly the missing basis elements, producing e_I up to a second Koszul sign. Because $(J \setminus I) \cap (I \setminus J) = \varnothing$, none of these steps kills the intermediate vector. With the chosen order, the total sign is the single scalar $\varepsilon_{I,J}$ built into the definition of $T_{I,J}$. Hence $T_{I,J}(e_{J'}) = 0$ for $J' \neq J$ and $T_{I,J}(e_J) = e_I$, so $T_{I,J} = E_{I,J}$.

The image of the split Clifford action therefore contains all matrix units of $\mathrm{End}(\Lambda W)$ and is surjective. Finally, $\dim \mathrm{Cl}(Q) = 2^{\dim(W^* \oplus W)} = 2^{2n}$ and $\dim \mathrm{End}(\Lambda W) = (2^n)^2 = 2^{2n}$, so the surjective map is an isomorphism. $\square$

Corollary 6.3 (Even part and half-spin consequences). *For the same split hyperbolic form $Q = \mathrm{dualProd}_K W$, the even Clifford image preserves the parity decomposition $\Lambda W = \Lambda^{\mathrm{even}} W \oplus \Lambda^{\mathrm{odd}} W$. On each parity block it contains all matrix units between basis vectors of the same parity. If* $\dim W > 0$*, then*

$$\mathrm{Cl}^+(Q) \simeq \mathrm{End}(\Lambda^{\mathrm{even}} W) \times \mathrm{End}(\Lambda^{\mathrm{odd}} W).$$

If $\dim W = 0$*, then* $\mathrm{Cl}^+(Q) = K = \mathrm{End}(\Lambda^{\mathrm{even}} W)$ *and* $\Lambda^{\mathrm{odd}} W = 0$*. Moreover,* $\Lambda^{\mathrm{even}} W$ *is simple as a* $\mathrm{Cl}^+(Q)$*-module, and if* $\dim W > 0$ *then* $\Lambda^{\mathrm{odd}} W$ *is simple as well. When* $\dim W > 0$ *the two parity pieces are inequivalent.*

Proof. Each L_i and D_i changes degree by ± 1, so even words preserve parity. If $|I| \equiv |J| \pmod 2$, then the operator $T_{I,J}$ above contains an even number of wedge/contraction steps and belongs to the even Clifford image. Restricting to basis vectors of fixed parity therefore yields every block matrix unit on $\Lambda^{\mathrm{even}} W$ and on $\Lambda^{\mathrm{odd}} W$. If $\dim W > 0$, this gives a surjective homomorphism

$$\mathrm{Cl}^+(Q) \twoheadrightarrow \mathrm{End}(\Lambda^{\mathrm{even}} W) \times \mathrm{End}(\Lambda^{\mathrm{odd}} W).$$

Now

$$\dim \mathrm{Cl}^+(Q) = 2^{2n-1},$$
$$\dim \mathrm{End}(\Lambda^{\mathrm{even}} W) + \dim \mathrm{End}(\Lambda^{\mathrm{odd}} W) = 2(2^{n-1})^2 = 2^{2n-1}.$$

so the surjection is an isomorphism. If $\dim W = 0$, then $\Lambda^{\mathrm{even}} W = K$, $\Lambda^{\mathrm{odd}} W = 0$, and the even Clifford algebra is just K.

The same blockwise matrix units show that any nonzero vector in $\Lambda^{\mathrm{even}} W$ generates the whole even half, and similarly for $\Lambda^{\mathrm{odd}} W$ when $n > 0$. When $n > 0$, any $\mathrm{Cl}^+(Q)$-linear map $\Lambda^{\mathrm{even}} W \to \Lambda^{\mathrm{odd}} W$ is annihilated by $0 \times \mathrm{End}(\Lambda^{\mathrm{odd}} W)$ on the source and acted on faithfully by that factor on the target, so it must vanish. $\square$

Proposition 6.4 (Transport to hyperbolic presentations and split rank). *If $e : Q \simeq \mathrm{dualProd}_K W$ is an explicit hyperbolic presentation, then conjugating the split-model operators above by the linear identification induced from e transports the matrix-unit theorem and its parity corollary to the Clifford action attached to Q. If Q is finite-dimensional, nondegenerate, and of split rank (equivalently maximal Witt index), a Witt decomposition provides such an e, so the same proof gives the split-rank statements without keeping the isometry as part of the final formulation.*

Proof. Let ρ_{hyp} denote the split-model Clifford action on ΛW, and let

$$\phi_e : \mathrm{Cl}(Q) \xrightarrow{\simeq} \mathrm{Cl}(\mathrm{dualProd}_K W)$$

be the algebra isomorphism induced by the hyperbolic presentation e. Transporting the action simply means defining

$$\rho_e(a) = \rho_{\mathrm{hyp}}(\phi_e(a)) \qquad (a \in \mathrm{Cl}(Q)).$$

For a vector $v \in V$, this says $\rho_e(\iota_Q(v)) = (\rho_{\mathrm{hyp}} \circ \iota_{\mathrm{dualProd}} \circ e)(v)$, so the transported action is obtained by writing the split-model wedge/contraction formulas in the coordinates supplied by e. Because ϕ_e is an algebra isomorphism, the transported wedge/contraction operators satisfy the same Clifford relations as the split ones. The projectors P_I and transfer operators $T_{I,J}$ are algebraic expressions in those operators, so the entire matrix-unit construction survives unchanged under transport. This gives the explicit-presentation version of both Theorem 6.2 and Corollary 6.3.

Now suppose Q is finite-dimensional, nondegenerate, and of split rank (equivalently maximal Witt index). Choose a maximal totally isotropic subspace $W \subset V$ and an isotropic complement W'. Witt decomposition gives $V = W' \oplus W$, and the bilinear pairing between W' and W identifies W' with W^*, producing a hyperbolic presentation $Q \simeq \mathrm{dualProd}_K W$. Applying the preceding transport to that presentation yields the split-rank statements. The final theorems depend only on the existence of such a Witt decomposition, not on retaining a particular choice in the statement. □

Identifying Weyl spinors without confusing them with the ambient regular module. One subtle point is that the ambient regular module on V has dimension $2^{\dim V}$, whereas the chosen spinor model in split rank has the expected dimension $2^{\dim V/2}$. The half-spin theorems therefore cannot literally be statements about ambient submodules of the regular representation. Instead, bridge results identify the canonical chosen exterior-model halves with the even and odd parts of the exterior model built on the maximal isotropic space. This is the formal content of the slogan $S^{\pm} = \Lambda^{\mathrm{even/odd}} W$.

Covering-map results: exact kernel, spinorial descent, explicit transvections, chosen-line square scaling, and exact split-line image. The formalized covering-map results isolate the exact field-sensitive pieces needed for the chosen exterior-model theorems, under the standing $2 \in K^{\times}$ hypothesis. In the split-rank setting treated here, if an element lies in the kernel of the spin-to-isometry representation, then its conjugation action is trivial on the Clifford generators, hence on the whole Clifford algebra. The matrix model then forces such an element to be scalar, and the spin norm reduces the scalar possibilities to ± 1.

Proposition 6.5 (Kernel calculation from the matrix model). *Assume Q is finite-dimensional, nondegenerate, and hyperbolic. If $x \in \mathrm{Spin}(Q)$ acts trivially on V under the spin-to-isometry map, then $x = \pm 1$.*

Proof. Triviality of the orthogonal action means $xvx^{-1} = v$ for every $v \in V$, so $xv = vx$. Since V generates $\mathrm{Cl}(Q)$, the element x is central in $\mathrm{Cl}(Q)$. Under the matrix-model isomorphism $\mathrm{Cl}(Q) \simeq \mathrm{End}(\Lambda W)$, central elements correspond to the center of a full matrix algebra, hence to scalars. Thus $x = \lambda$ for some $\lambda \in K^{\times}$. The defining spin norm relation then gives $1 = x\tilde{x} = \lambda^2$, so $\lambda = \pm 1$. Transporting this argument through the split-Witt reformulation gives the positive split-rank kernel theorem used in the main results. $\square$

Proposition 6.6 (Positive split-rank non-factorization). *Assume Q is finite-dimensional, nondegenerate, and of positive split rank, and let $S = \Lambda W$ be the exterior-model spinor module attached to a chosen Witt decomposition of Q. Then the spin representation*

$$\mathrm{Spin}(V, Q) \to \mathrm{GL}(S)$$

does not factor through $\mathrm{SO}(V, Q)$. The same is true for the induced representations on the nonzero half-spin modules S^+ and S^-.

Proof. Choose a nonzero $w \in W$ and a dual vector $f \in W^*$ with $f(w) = 1$. In the transported hyperbolic model, the vector

$$u = (-f, w) \in W^* \oplus W$$

satisfies

$$Q(u) = (-f)(w) = -1,$$

hence $u^2 = -1$. Therefore

$$-1 = u^2$$

is a product of two unit vectors, so $-1 \in \mathrm{Spin}(V, Q)$. Being scalar, it acts trivially on V by conjugation, so its image in $\mathrm{SO}(V, Q)$ is the identity. On the spinor module $S = \Lambda W$, however, the Clifford action of the scalar -1 is just $-\mathrm{id}_S$, which is nontrivial because $(-1) \cdot 1 = -1 \neq 1$ in S. Since scalar multiplication preserves parity, the same operator restricts to $-\mathrm{id}$ on each half-spin piece. The positive half contains 1, and the negative half is nonzero because $w \in W \subset \Lambda^1 W$. Thus the action of -1 is nontrivial on S, S^+, and S^-. Therefore none of these representations can factor through $\mathrm{SO}(V, Q)$. $\square$

Proposition 6.7 (Positive split-rank projective descent). *Assume Q is finite-dimensional, nondegenerate, and of positive split rank, and let $S = \Lambda W$ be the exterior-model spinor module attached to a chosen Witt decomposition of Q. Then the induced action of $\mathrm{Spin}(V, Q)$ on the projective space $\mathbb{P}(S)$ of one-dimensional subspaces depends only on the image of $\mathrm{Spin}(V, Q) \to \mathrm{SO}(V, Q)$ and therefore defines an action of that image subgroup. The same is true for the projective spaces of the nonzero half-spin modules S^+ and S^-. In Lean this is packaged as an explicit action of the spin-to-isometry image subgroup on chosen exterior-model submodules; the projective statement is the restriction to one-dimensional submodules.*

Proof. Suppose $x, y \in \mathrm{Spin}(V, Q)$ have the same image in $\mathrm{SO}(V, Q)$. Then $y^{-1}x$ acts trivially on V, so by the kernel calculation above one has $y^{-1}x = \pm 1$. On S, and likewise on each half-spin piece, the scalars ± 1 act by scalar multiplication and therefore fix every one-dimensional subspace. Thus x and y induce the same transformation on $\mathbb{P}(S)$, $\mathbb{P}(S^+)$, and $\mathbb{P}(S^-)$. Hence these projective actions depend only on the image in $\mathrm{SO}(V, Q)$ and therefore define actions of the image subgroup of $\mathrm{Spin}(V, Q) \to \mathrm{SO}(V, Q)$. $\square$

Proposition 6.8 (Explicit hyperbolic transvection Clifford unit). *Let* $Q = \mathrm{dualProd}_K W$. *If* $\delta \in W^*$ *and* $w \in W$ *satisfy* $\delta(w) = 0$, *set* $a = (\delta, 0)$, $b = (0, w)$, *and* $x_{\delta,w} = 1 + \iota(a)\iota(b) \in \mathrm{Cl}(Q)$. *Then* $x_{\delta,w}$ *is an even Clifford unit with*

$$x_{\delta,w}^{-1} = 1 - \iota(a)\iota(b),$$

and for every $(d, u) \in W^* \oplus W$ *one has*

$$x_{\delta,w}\, \iota(d, u)\, x_{\delta,w}^{-1} = \iota(d + d(w)\delta,\ u - \delta(u)w).$$

Equivalently, conjugation by $x_{\delta,w}$ *realizes the transported hyperbolic transvection attached to* (δ, w).

Proof. Write $n = \iota(a)\iota(b)$. Because $Q(a) = Q(b) = 0$ and $\langle a, b\rangle = \delta(w) = 0$, one has $n^2 = 0$. Hence

$$(1 + n)(1 - n) = 1 = (1 - n)(1 + n),$$

so $x_{\delta,w}^{-1} = 1 - n$. The element n is a product of two vectors, hence lies in the even Clifford part, and therefore so does $x_{\delta,w}$.

Now let $z = (d, u)$. Using the Clifford anticommutation relation in the hyperbolic form,

$$\iota(b)\iota(z) + \iota(z)\iota(b) = \langle b, z\rangle = d(w), \qquad \iota(z)\iota(a) + \iota(a)\iota(z) = \langle z, a\rangle = \delta(u),$$

one obtains

$$n\, \iota(z) = d(w)\iota(a) - \iota(a)\iota(z)\iota(b), \qquad \iota(z)\, n = \delta(u)\iota(b) - \iota(a)\iota(z)\iota(b).$$

Since $n^2 = 0$, also $n\, \iota(z)\, n = 0$, and therefore

$$\begin{aligned} x_{\delta,w}\, \iota(z)\, x_{\delta,w}^{-1} &= (1 + n)\iota(z)(1 - n) \\ &= \iota(z) + (n\iota(z) - \iota(z)n) \\ &= \iota(d + d(w)\delta,\ u - \delta(u)w). \end{aligned}$$

This is exactly the displayed transvection formula. □

Unipotent and semisimple directions in the hyperbolic model. Proposition 6.8 gives an explicit Clifford-unit representative for the unipotent root-subgroup direction in the hyperbolic Levi picture. The next proposition provides the complementary semisimple direction along a chosen split line.

Proposition 6.9 (Chosen-line square scaling in the hyperbolic model). *Let* $Q = \mathrm{dualProd}_K W$. *Choose* $w \in W$ *and* $f \in W^*$ *with* $f(w) = 1$. *For* $t \in K^\times$, *define*

$$\lambda_t(d, u) = (d + (t^{-2} - 1)d(w)f,\ u + (t^2 - 1)f(u)w).$$

Then λ_t *scales the line* Kw *by* t^2, *scales the dual line* Kf *by* t^{-2}, *fixes* $\ker(f) \subset W$ *and* $\ker(\mathrm{ev}_w) \subset W^*$, *and lies in the image of* $\mathrm{Spin}(Q) \to \mathrm{SO}(Q)$. *More precisely, if*

$$u_t = (-t^{-1}f, tw), \qquad v = (-f, w),$$

then $Q(u_t) = Q(v) = -1$ *and the spin element* $u_t v$ *acts on* $W^* \oplus W$ *as* λ_t.

Proof. Because $f(w) = 1$,

$$Q(u_t) = (-t^{-1}f)(tw) = -1, \qquad Q(v) = (-f)(w) = -1,$$

so u_t and v are vectors of square -1, hence $u_t v \in \mathrm{Spin}(Q)$. A direct reflection calculation gives, for any $c \in K^\times$,

$$r_{(-c^{-1}f,cw)}(d,u) = ((d(w) - c^{-2}f(u))f - d,\ -(c^2 d(w) - f(u))w - u).$$

Applying first the reflection with $c = 1$ and then the reflection with $c = t$ yields

$$(d,u) \longmapsto (d + (t^{-2} - 1)d(w)f,\ u + (t^2 - 1)f(u)w) = \lambda_t(d,u).$$

The displayed formula makes the scaling and fixed-kernel claims immediate, so λ_t is a square scaling along the chosen split line and lies in the spin image. $\square$

If $\ell_t \in \mathrm{GL}(W)$ denotes the underlying line scaling

$$\ell_t(u) = u + (t^2 - 1)f(u)w,$$

then the same explicit lift has a fully normalized exterior-model action:

$$\rho_S(u_t v) = -t^{-1}\,\mathrm{Ext}(\ell_t).$$

This is the exact torus-lift refinement of the projective Levi-action theorem, formalized as `splitCliffordAction_spinIotaPairOfQuadraticEqNegOne_eq_smul_exteriorMap_lineScaling`.

Corollary 6.10 (Chosen-line torus control of the transvection parameter). *Fix $w \in W$ and $f \in W^*$ with $f(w) = 1$. Let λ_t be the square-scaling map from Proposition 6.9, and for $\delta(w) = 0$ let $T_{\delta,w}$ denote the transvection from Proposition 6.8,*

$$T_{\delta,w}(d,u) = (d + d(w)\delta,\ u - \delta(u)w).$$

Then

$$\lambda_t\, T_{\delta,w}\, \lambda_t^{-1} = T_{t^2\delta,w}.$$

Consequently the explicit lifts in Propositions 6.8 and 6.9 control both the unipotent root subgroup

$$U_w = \{\, T_{\delta,w} : \delta(w) = 0 \,\}$$

*and the square torus generated by the λ_t, with the expected weight-*2 *conjugation action.*

Proof. Write $d = d_0 + d(w)f$ with $d_0(w) = 0$ and $u = u_0 + f(u)w$ with $f(u_0) = 0$. Then

$$\lambda_t(d,u) = (d_0 + t^{-2}d(w)f,\ u_0 + t^2 f(u)w),$$

so λ_t^{-1} is obtained by replacing t with t^{-1}. If $\delta(w) = 0$, then λ_t fixes δ and scales w by t^2. Applying λ_t^{-1}, then $T_{\delta,w}$, and then λ_t gives

$$(d,u) \longmapsto (d + t^2 d(w)\delta,\ u - t^2\delta(u)w) = T_{t^2\delta,w}(d,u),$$

which is the displayed conjugation formula. The final claim is just a reformulation: the manuscript gives explicit Clifford-unit representatives for the unipotent subgroup U_w and an explicit spin lift of the semisimple square torus acting on it. □

Corollary 6.11 (Internal Clifford-level torus action). *Fix $w \in W$ and $f \in W^*$ with $f(w) = 1$, and let*

$$s_t = \iota(-t^{-1} f, tw)\, \iota(-f, w) \in \mathrm{Spin}(\mathrm{dualProd}_K W)$$

be the explicit chosen-line square-scaling lift from Proposition 6.9. For $\delta(w) = 0$, the explicit transvection units satisfy

$$s_t\, x_{\delta,w}\, s_t^{-1} = x_{t^2\delta,w}.$$

Thus the weight-2 torus action already holds inside the explicit Clifford representatives, not only after passing to the orthogonal action.

Proof. By Proposition 6.9, conjugation by s_t fixes the vector $(\delta, 0)$ and sends $(0, w)$ to $(0, t^2 w)$. Therefore

$$s_t\, x_{\delta,w}\, s_t^{-1} = 1 + \iota(\delta, 0)\, \iota(0, t^2 w).$$

Because ι is K-linear and scalars are central in the Clifford algebra,

$$\iota(\delta, 0)\, \iota(0, t^2 w) = \iota(\delta, 0)\, t^2 \iota(0, w) = \iota(t^2\delta, 0)\, \iota(0, w),$$

so the right-hand side is exactly $x_{t^2\delta,w}$. □

Corollary 6.12 (Semidirect product along a chosen split line). *For fixed $w \in W$ and $f \in W^*$ with $f(w) = 1$, the maps $T_{\delta,w}$ with $\delta(w) = 0$ satisfy*

$$T_{\delta,w}\, T_{\eta,w} = T_{\delta+\eta,w} \qquad (\delta(w) = \eta(w) = 0),$$

so they form an additive subgroup naturally identified with $\ker(\mathrm{ev}_w) \subset W^*$. Together with the square scalings λ_t, the manuscript therefore produces the semidirect-product structure

$$\ker(\mathrm{ev}_w) \rtimes K^\times, \qquad t \cdot \delta = t^2\delta,$$

inside the orthogonal action along the chosen line Kw.

Proof. Apply the explicit formula for $T_{\eta,w}$ first:

$$(d, u) \longmapsto (d + d(w)\eta,\ u - \eta(u)w).$$

Because $\eta(w) = 0$, the scalar $d(w)$ is unchanged on the dual side, and $\delta(u - \eta(u)w) = \delta(u)$ because $\delta(w) = 0$. Therefore

$$T_{\delta,w} T_{\eta,w}(d, u) = (d + d(w)(\delta + \eta),\ u - (\delta(u) + \eta(u))w) = T_{\delta+\eta,w}(d, u).$$

This proves the additive subgroup law. The semidirect-product statement is then the combination of this identity with Corollary 6.10, which shows $\lambda_t T_{\delta,w} \lambda_t^{-1} = T_{t^2\delta,w}$. □

The previous results supply the explicit root and torus directions. We now prove the structural Levi-action theorem from Section 5.

Proof of Theorem 5.3. Let $v = \rho_S(s)1 \in S$. For any $d \in W^*$, because s projects to $\Lambda(g)$, conjugation carries the dual generator $\iota(g^\vee d, 0)$ to $\iota(d, 0)$. Applying both sides to 1 gives

$$\iota(d, 0)\, v = \rho_S(s)(\iota(g^\vee d, 0) \cdot 1) = 0,$$

since dual generators act on S by contraction and annihilate the vacuum vector 1. Thus every contraction operator kills v, and the vacuum-line criterion from Section 4 (formalized as `eq_algebraMap_of_forall_contractionAction_eq_zero`) gives $v = c \cdot 1$ for some $c \in K$.

Similarly, for every $w \in W$, conjugation carries $\iota(0, w)$ to $\iota(0, gw)$, so $\rho_S(s)$ intertwines exterior multiplication by w with exterior multiplication by gw. Starting from $\rho_S(s)1 = c$ and inducting on exterior monomials yields

$$\rho_S(s) = c\ \mathrm{Ext}(g).$$

Because both $\rho_S(s)$ and $\mathrm{Ext}(g)$ are invertible, necessarily $c \neq 0$; the corresponding Lean theorem records this final scalar directly as an element of $K^\times$. □

The preceding theorem is structural: it identifies the chosen exterior-model action of any split Levi lift up to scalar. We now prove the corresponding finite-basis factorization, explicit-lift, and exact image theorem from Section 5.

Proof of Theorem 5.4. Write the matrix of g in the chosen basis. If $n = 1$, the factorization has empty transvection lists and no 2×2 blocks; the determinant-square hypothesis gives exactly the chosen-line square scaling. Assume now that $n \geq 2$. The classical transvection reduction for invertible matrices in dimension at least 2 [1, Ch. IV] writes g as basis transvections, a diagonal basis scaling, and basis transvections. Because $\det(g)$ is a square, the middle diagonal factor has square total determinant. The basis-diagonal square-determinant theorem proved in the formal development therefore splits that middle factor as one chosen-line square scaling together with the canonical determinant-one 2×2 diagonal blocks supported on the distinguished basis line and each remaining basis line.

Transporting to $\mathrm{SO}(\mathrm{dualProd}_K W)$ yields exactly the displayed basis-dependent factorization of $\Lambda(g)$. Each transported basis transvection is realized by the explicit Clifford unit from Proposition 6.8, while each canonical 2×2 block factors further as four transported hyperbolic transvections. Hence $\Lambda(g)$ is indeed a product of transported hyperbolic transvections and one chosen-line square scaling, and every factor is implemented by the explicit Clifford formulas already constructed above. This is the claimed orthogonal factorization.

Applying the exterior-model lift theorem to the same decomposition gives an even unitary Clifford unit x built from the left transvection list, the square-determinant diagonal factor, and the right transvection list. The transvection lists and the determinant-one diagonal block product act exactly by the corresponding exterior maps, while the chosen-line square scaling contributes the scalar factor $-u^{-1}$. Multiplying those Clifford units therefore yields

$$\rho_S(x) = -u^{-1}\,\mathrm{Ext}(g),$$

which is the claimed explicit exterior-model lift. For the square-surjective corollary, apply surjectivity of $(K^\times) \xrightarrow{(-)^2} (K^\times)$ to the unit $\det(g)$, choose the resulting u, and invoke the same square-determinant lift theorem.

For the spin-image sufficiency direction, the formal proof replaces the auxiliary third-basis-direction argument with a four-vector Clifford factorization of every basis transvection unit. Thus each transported basis transvection is already a spin element in finite rank, without a rank-at-least-3 restriction. Products of these transvection spin lifts, together with the chosen-line square-scaling spin lift, realize the same square-determinant factorization inside the image of $\mathrm{Spin}(\mathrm{dualProd}_K W) \to \mathrm{SO}(\mathrm{dualProd}_K W)$.

For the converse, let a spin element s project to $\Lambda(g)$. The Levi-projective exterior-action theorem gives $\rho_S(s) = c\,\mathrm{Ext}(g)$ for some $c \in K^\times$. The top exterior pairing

$$B(a, b) = \mathrm{top}(\mathrm{reverse}(a)b)$$

is adjoint to the split Clifford action with respect to Clifford reversion, and for spin elements the reversion $s^\star$ agrees with the Clifford conjugation $\tilde{s}$ fixed in the standing conventions. Since $s^\star s = 1$, this adjointness forces the bottom-line scalar and the top exterior coefficient of $\mathrm{Ext}(g)$ to multiply to 1. But the latter top coefficient is $\det(g)$, so the normalization gives $\det(g) = c^{-2}$, a square in $K^\times$. This proves the iff. The same square-surjectivity argument then removes the determinant-square hypothesis for all finite-basis Levi elements with a distinguished basis line. Surjectivity of the square-class character is proved by the complementary diagonal calculation: scaling the distinguished basis line by any unit has that unit as its determinant and hence realizes the chosen square class. The formal quotient equivalence is then the standard first-isomorphism-theorem map for this surjective character, in both linear coordinates and on the transported split Levi, after replacing the relevant kernel by the proved spin-image subgroup. The exported spinor-norm-facing declarations are local aliases for this finite-basis split-Levi character and its quotient equivalence. □

A concrete rank-2 shear. If $W = K^2$ with basis e_1, e_2, take $w = e_1$, $f = e_1^*$, and $\delta = e_2^*$. Then Proposition 6.8 gives the explicit higher-rank shear

$$(d_1, d_2, u_1, u_2) \longmapsto (d_1, d_2 + d_1,\ u_1 - u_2,\ u_2).$$

Corollary 6.10 says that the chosen-line square scalings rescale this shear parameter by t^2, exhibiting the standard root-group / torus interaction already inside the explicit Clifford formulas.

Proposition 6.13 (Split-line square-scaling calculation). *Let $Q = \mathrm{dualProd}_K K$, and choose isotropic generators p and q with $p^2 = q^2 = 0$ and $pq+qp = 1$. Every even Clifford element can then be written as $x = a\,pq + b\,qp$. If $x \in \mathrm{Spin}(Q)$, the* *defining norm condition forces $ab = 1$, and an explicit conjugation calculation gives*

$$xpx^{-1} = a^2 p, \qquad xqx^{-1} = a^{-2} q.$$

Hence the image of $\mathrm{Spin}(Q) \to \mathrm{SO}(Q)$ *is exactly the square-scaling subgroup.*

Proof. Because the even part is spanned by pq and qp, every even element has the stated form. Using the relations $p^2 = q^2 = 0$ and $pq + qp = 1$, one has

$$(pq)^2 = pq, \qquad (qp)^2 = qp, \qquad pq\,qp = qp\,pq = 0.$$

The Clifford conjugate of pq is qp. Hence

$$\tilde{x} = a\,qp + b\,pq,$$

so

$$x\tilde{x} = (a\,pq + b\,qp)(a\,qp + b\,pq) = ab(pq + qp) = ab.$$

Thus the spin condition $x\tilde{x} = 1$ is equivalent to $ab = 1$, and in that case $x^{-1} = \tilde{x}$. Moreover,

$$xp = (a\,pq + b\,qp)p = a\,p, \qquad xq = (a\,pq + b\,qp)q = b\,q,$$

because $pqp = p$, $qpp = 0$, $pqq = 0$, and $qpq = q$. Therefore

$$xpx^{-1} = a\,p(a\,qp + b\,pq) = a^2 p, \qquad xqx^{-1} = b\,q(a\,qp + b\,pq) = b^2 q = a^{-2} q.$$

Hence the induced orthogonal action rescales the two isotropic lines by reciprocal squares, so the image is contained in the square-scaling subgroup. Conversely, Proposition 6.9 specializes to the split line and shows that every square-scaling occurs in the image. This is exactly the square-scaling subgroup. □

This shows that naive surjectivity is false over fields with nonsquare units, and it pinpoints the extra hypothesis needed to recover the expected statement: if the square map on $K^\times$ is surjective, then one recovers the full split-line double cover. Combined with the positive split-rank linear/projective descent picture, it shows that the exterior-model spin action has a sharp two-level structure over a general field: projectively it depends only on the orthogonal image in every positive split rank, while linearly it remembers the scalar kernel element -1; in rank 1, the image can additionally miss nonsquare reciprocal scalings.

7. Conclusion

Starting from the classical exterior-model construction, we prove a field-sensitive covering/image theory. Its conceptual hinge is the theorem that split Levi lifts act projectively as the natural exterior action on ΛW. The explicit transvection and chosen-line square-scaling lifts feed into that bridge, and the finite-basis square-determinant theorem produces both an orthogonal factorization and an explicit even unitary Clifford lift.

The top exterior pairing and spin normalization prove the converse determinant-square obstruction. Thus finite-basis split-Levi spin-image membership is exactly the square-determinant condition. Equivalently, the split-Levi square-class character is onto with the spin image as kernel, so the split-Levi quotient by the spin image is the square-class group. Over fields with square-surjective unit group, every finite-basis Levi element lies in the spin image; algebraically closed fields are a standard instance of this square-surjective case.

The product-level Clifford norm formula is also available with order invariance, repeated-pair cancellation, and duplicated-product/subproduct triviality for selected vector lists. The present results stop at the product-level Clifford norm formula and the finite-basis split-Levi image theorem; global orthogonal-group spinor norms, full orthogonal-image classification, full Bott-periodic classification, and scheme-theoretic spin covers remain outside the scope of this paper.

Statements and Declarations

Funding. No funding was received to assist with the preparation of this manuscript.
Competing interests. The authors have no competing interests to declare that are relevant to the content of this article.
Data availability. No datasets were generated or analysed during the current study.
Code availability. The formal verification source code supporting the results of this article is available at `https://github.com/Arthur742Ramos/SpinorLean`; the code artifact referenced by this paper is source revision `e7c7bf09e6c4477124e85a7d4339c0b4a97e52ca`.

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